CSC1029 OO Programming

Lecture 13 – Preparation for Assessment 1

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What's to come in today's lecture?

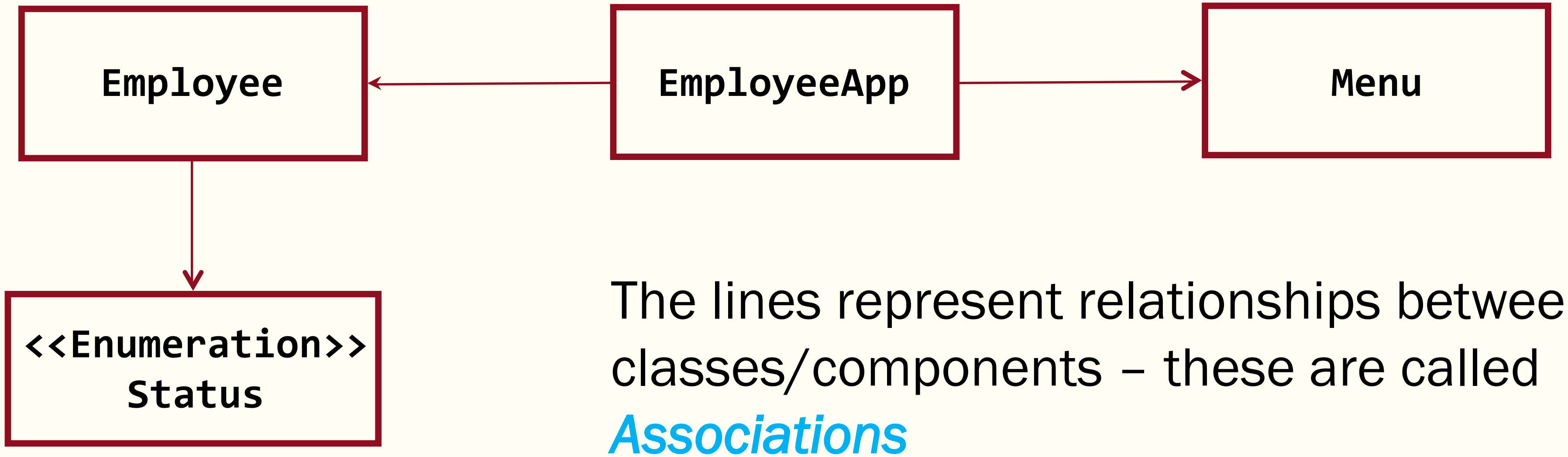
- How to Prepare for Assessment 1
- Review of a UML Problem Specification
- Implementation in Java
- How to Submit to Canvas

Assessment 1 Details

- Worth 20% of the module
- A Practical Exam – taken in the labs
- Wednesday 15th February 2023
- 1 Hour and 20 minutes in duration
- Extra time for ISSA students (TBA)

UML for an Employee Application (1)

Here we have a partial UML diagram for an Employee Management Application



The lines represent relationships between classes/components – these are called *Associations*

We will describe the different types of association in the coming lectures.
What do you think the arrows indicate?

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UML for an Employee Application (2)

```
<<Enumeration>>
Status
---------------
FULL_TIME
PART_TIME
LEFT
```

The Employee Management Application should use the Menu class to provide the following options to the user:

```
Employee Management
+++++++++++++++++++

1. Add Employee
2. Select Employee
3. List All
4. Quit

Enter choice:
```

```
Employee
---------------------------------
- empNum: int
- nextNum: int = 1
- name: String
- address: String
- salary: double
- eStat: Status
---------------------------------
+ Employee(String, String)
+ Employee(String, String,
               double)
+ getDetails(): String
+ toString(): String
+ calcGrossWage(): double
+ calcNetWage(): double
+ calcTax(): double
```

Code Implementation (1)

Part 1

Implementing the *Enumeration*
Implementing the *Employee class*
Implementing a *tester application*

Code Implementation (2)

Part 2

Implementing an array list of *Employee* objects

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Code Implementation (3)

Part 3

Implementing the *main application*

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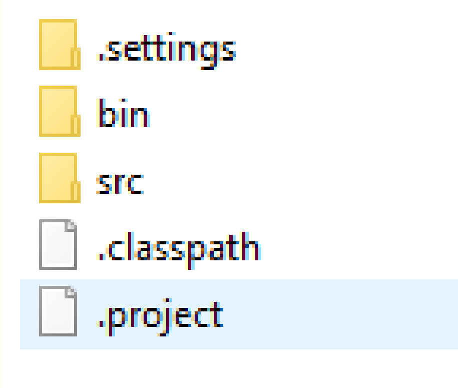
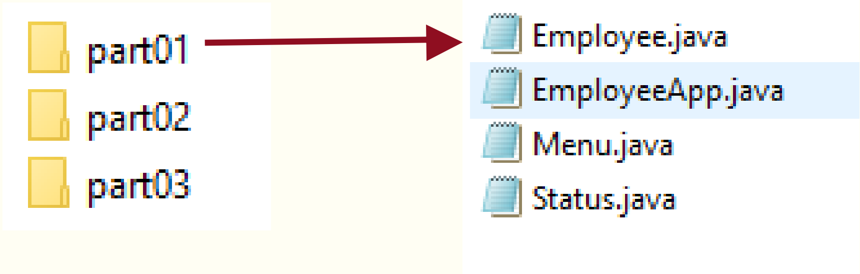
Submitting to Canvas

Archive your ***source*** code (.java files)

- Navigate to your ***project*** folder

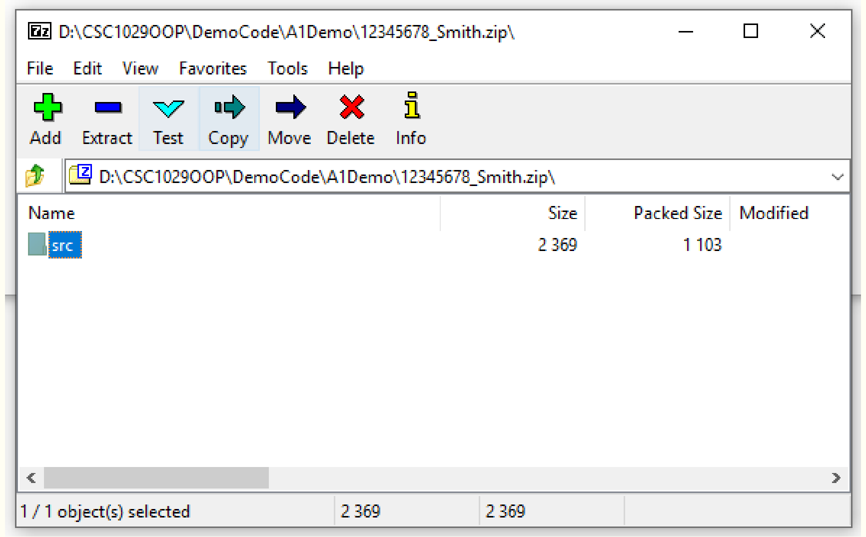


- Open the ***src*** folder



- right-click src folder
- send to > Compressed (zip) folder
- ***rename*** the zip file:

  e.g 12345678_Smith.zip
- ***check*** zip contents:



- follow instructions to upload

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Javadoc and Other Comments (1)

An overall comment (Javadoc) for the class.

```java
import java.util.Scanner;

/**
 * This class represents ...
 * @author Paul Sage
 * @version V1.0
 *
 */
public class Menu {
```

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What's Next?

Class Associations

Aggregation & Composition